

RED TEAM LEARNING PATH

Module 23

Command & Control Concepts

Architecture, Tasking, Telemetry, Detection, and a Controlled Local C2 Simulator

Module goal - Memahami C2 sebagai sistem komunikasi terkontrol antara operator dan endpoint, lalu mampu menjelaskan lifecycle, transport, tasking, telemetry, detection opportunities, dan operational hygiene tanpa bergantung pada framework tertentu.

Safety boundary - Seluruh hands-on wajib menggunakan simulator localhost yang disertakan atau lab yang secara eksplisit authorized. Modul ini tidak mengajarkan AV/EDR bypass, covert redirector infrastructure, domain fronting, stealth tuning, persistence, credential theft, atau deployment agent ke sistem pihak lain.

Outcome	Setelah module ini lo harus bisa...
Architecture	Menggambar operator -> team server -> transport -> agent serta menjelaskan trust boundary setiap komponen.
Lifecycle	Menjelaskan enrollment/check-in, task queue, execution, result return, artifact transfer, audit, dan teardown.
Protocols	Menghubungkan HTTP(S), DNS, mTLS/WireGuard dan channel lain ke ATT&CK T1071 secara konseptual.
Telemetry	Membaca process/network/DNS logs dan mengenali periodic callback, unusual process-to-network mapping, serta transfer artifact.
Framework literacy	Mereview Sliver secara aman: provenance, architecture, transports, operator model, dan help/reference tanpa membuat implant.
Defensive thinking	Menulis detection hypothesis, expected telemetry, cleanup checklist, dan capstone report.

Prerequisites

- Module 00-22, terutama networking, HTTP, Windows/Linux fundamentals, ATT&CK, evidence handling, dan detection-aware operations.
- Omarchy/Arch Linux dengan Python 3; optional: tcpdump/Wireshark/minisign.
- Untuk Windows telemetry: disposable Windows VM + Sysmon jika tersedia.

Tooling di Omarchy

omarchy pkg add python jq tcpdump wireshark-qt minisign

Package tersebut tersedia di repository resmi Arch; untuk Sliver, modul ini memakai jalur review-only dan menyarankan binary resmi yang diverifikasi signature, bukan random AUR build.

1. C2 sebagai Sistem, Bukan Sekadar "Remote Shell"

Command and Control (C2) adalah lapisan komunikasi dan koordinasi setelah sebuah endpoint berada dalam konteks operasi yang authorized. Remote shell hanya salah satu capability; C2 yang matang biasanya juga menangani identity agent, task queue, transport state, result collection, file/artifact handling, multi-operator access, audit trail, dan lifecycle cleanup.

Komponen	Peran	Pertanyaan security
Operator	Manusia/console yang memberi task dan membaca hasil.	Siapa boleh tasking? Apakah action ter-audit?
Team server	State pusat: agents, tasks, results, listeners, operator auth.	Secrets di mana? Bind address? TLS? Audit log?
Listener / transport	Channel yang menerima check-in atau session.	Protocol apa? Port apa? Egress rule apa?
Agent / beacon	Endpoint component yang check-in dan menjalankan capability.	Privilege? Parent process? Network identity? Cleanup?
Task queue	Buffer command/capability yang menunggu agent mengambil task.	Idempotency? Expiry? Operator attribution?
Result / loot store	Output, files, metadata, screenshots, dll.	Sensitive data? Encryption at rest? Retention?

1.1 Session vs Beacon

Model	Karakter	Operational trade-off	Detection clue
Interactive session	Channel relatif kontinu / near-real-time.	Responsif tetapi koneksi terlihat lebih lama.	Long-lived connection, process-network correlation.
Beacon / polling	Agent periodik check-in, ambil task, kirim result.	Lebih asynchronous; operator tidak selalu real-time.	Periodic timing, repeated destination, small request/response pattern.

Think like a defender - Jangan mencari string "C2". Cari hubungan perilaku: proses mana yang membuat koneksi, ke mana, seberapa periodik, ukuran data, DNS yang dipakai, apa yang terjadi sebelum/sesudah koneksi, dan apakah pola tersebut wajar untuk host itu.

1.2 Lifecycle C2

Provision -> Listener -> Agent enrollment -> Check-in -> Tasking -> Execution -> Result return -> Artifact transfer -> Monitoring -> Teardown

- Provision: server/operator identity, certificates, bind addresses, storage, logging.
- Enrollment: agent memperoleh identifier dan server mengaitkan metadata endpoint.
- Check-in: transport membawa heartbeat/status dan mengambil task.
- Tasking: operator memilih capability; server menyimpan siapa mengirim task dan kapan.
- Execution: endpoint menjalankan capability dalam privilege/context tertentu.
- Result return: stdout/structured result/artifact dikirim balik.

- Teardown: stop listener, terminate agent, remove lab artifacts, revoke credentials, archive evidence.

1.3 Trust boundaries yang wajib dipikirkan

Boundary	Jika gagal	Control
Operator -> server	Unauthorized tasking / credential theft.	Strong auth, least privilege, operator-specific identity.
Server -> agent	Agent menerima task dari pihak salah.	Authenticated transport, key/cert validation.
Agent -> OS	Capability berjalan terlalu privileged.	Least privilege, scoped capabilities, lab containment.
Server -> evidence store	Loot bocor atau tertukar.	Access control, encryption, hashing, retention policy.

2. C2 Transport dan MITRE ATT&CK

MITRE ATT&CK menempatkan Application Layer Protocol pada T1071. Sub-technique mencakup Web Protocols, File Transfer Protocols, Mail Protocols, DNS, dan Publish/Subscribe Protocols. T1105 Ingress Tool Transfer mencakup pemindahan tool/file ke environment target. Di module ini kita memakai konsep tersebut untuk memahami observability, bukan untuk menyamarkan traffic.

Transport / konsep	Apa yang dibawa	Defender melihat apa	Catatan lab
HTTP(S)	Check-in, task, result, metadata.	Process -> TCP 80/443, hostname, URI, TLS metadata, timing.	Simulator memakai HTTP plaintext di loopback agar mudah dianalisis.
DNS	Data kecil dalam query/response.	Process DNS event, unusual subdomain length/frequency, resolver path.	Konsep saja; tidak membuat DNS tunnel.
mTLS	Mutual-authenticated encrypted channel.	TLS connection + process/destination correlation.	Dibahas dari sisi trust/authentication.
WireGuard/VPN-like	Overlay channel antar endpoint/server.	Interface/routes/UDP flow/peer metadata.	Konsep; tidak membuat covert overlay.
File transfer	Artifact/payload/config/result.	Network download + file creation + hash + subsequent execution.	Lab hanya transfer file training fixed.

2.1 Beacon periodicity dan timing

Polling agent yang menghubungi server setiap interval dapat menghasilkan pola temporal. Detection tidak boleh hanya memakai satu threshold: updater, monitoring agent, browser, dan enterprise software juga bisa periodik. Kuncinya adalah menggabungkan timing dengan process identity, destination novelty, bytes, DNS, parent process, dan host role.

Signal	Weak sendiri karena...	Lebih kuat jika dikombinasikan dengan...
Interval stabil	Banyak software melakukan polling.	Unknown binary + rare destination + consistent small payloads.
HTTPS outbound	Hampir semua aplikasi modern memakai TLS.	Process tidak lazim + newly seen domain + no user interaction.
DNS cadence	Normal apps juga query DNS.	Long/entropy-heavy labels + process attribution + repeated failures.
Small POSTs	Telemetry/analytics juga kecil.	Periodic sequence + same endpoint + response/task behavior.

No stealth tuning here - Kita akan mengukur periodicity dan variasi traffic untuk tujuan detection. Modul ini sengaja tidak memberi recipe untuk jitter/traffic shaping/redirector yang bertujuan menghindari deteksi.

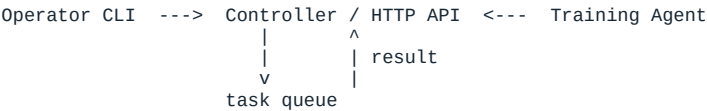
2.2 Telemetry map

Layer	Linux / Omarchy	Windows / Sysmon	Pertanyaan
Process	ps, /proc, journal	Event ID 1 ProcessCreate	Process apa yang memulai koneksi? Parent-nya siapa?
Network	ss, tcpdump, Wireshark	Event ID 3 NetworkConnect	Destination, port, protocol, process GUID?
DNS	tcpdump/Wireshark/system resolver logs	Event ID 22 DNS Query	Domain baru? Query periodik? Process mana?
File	find/stat/hash	Event ID 11 FileCreate	Ada artifact baru setelah network activity?

3. Safe Local C2 Simulator

Hands-on utama module ini memakai simulator Python yang sengaja dibatasi ke localhost. Agent tidak menerima arbitrary shell command. Task hanya berasal dari allow-list: system_info, process_summary, network_summary, echo, dan get_training_artifact. Tujuannya mempelajari arsitektur dan telemetry tanpa menghasilkan general-purpose implant.

127.0.0.1:8230



Allowed tasks only; no shell, no persistence, no remote host.

Endpoint	Method	Fungsi
/register	POST	Agent mendaftar metadata training.
/poll?id=...	GET	Agent mengambil satu queued task.
/result	POST	Agent mengirim structured result.
/artifact/training.txt	GET	Fixed harmless artifact untuk transfer lab.
/state	GET	Operator melihat agents/tasks/results.

3.1 Workspace

Lab 01 - Initialize C2 workspace

Objective: Membuat struktur evidence yang konsisten.

```
cd modules/23-command-control-concepts/labs
./workspace-init.sh module23-c2
```

- Pastikan folder logs, captures, artifacts, notes, evidence, dan results terbentuk.

Verify: Permission workspace sensitif seharusnya tidak world-readable.

3.2 Start controller dan agent

Lab 02 - Start controller

Objective: Menjalankan team-server simulator di loopback saja.

```
python c2-controller.py \
  --workspace module23-c2
```

- Controller harus menampilkan listening address 127.0.0.1:8230.

Verify: `ss -ltnp | grep 8230` hanya boleh menunjukkan 127.0.0.1.

Lab 03 - Start training agent

Objective: Mempelajari registration dan polling.

```
python c2-agent.py \
  --workspace module23-c2 \
  --agent-id lab-agent-01
```

- Biarkan terminal agent aktif; agent hanya menghubungi controller loopback.

Verify: Controller log harus mencatat register dan poll event.

3.3 Tasking tanpa shell

Lab 04 - Queue read-only tasks

Objective: Melihat task queue dan result lifecycle tanpa arbitrary command execution.

```
python operator.py system_info --agent lab-agent-01
python operator.py process_summary --agent lab-agent-01
python operator.py network_summary --agent lab-agent-01
python operator.py state
```

- Bandingkan queued -> delivered -> result timestamps.

Verify: Tidak ada task bernama shell/exec/run-command. Kalau lo menambahkan capability sendiri, tetap pertahankan allow-list dan lab boundary.

4. Observe Traffic, Timing, dan Task Flow

Lab 05 - Capture loopback traffic

Objective: Melihat HTTP request/response yang membentuk check-in.

```
sudo tcpdump -i lo -nn \
  tcp port 8230 \
  -w module23-c2/captures/c2-local.pcap
```

- Queue beberapa task lalu hentikan capture dengan Ctrl-C.

Verify: PCAP harus hanya berisi traffic lab pada port 8230.

Lab 06 - Inspect with Wireshark

Objective: Menghubungkan endpoint HTTP dengan lifecycle C2.

- Buka c2-local.pcap di Wireshark.
- Gunakan display filter: `tcp.port == 8230`
- Cari /register, /poll, /result, dan /artifact/training.txt.
- Catat request size, response size, method, interval, dan urutan state.

Verify: Bisa menjelaskan packet flow tanpa membaca source code simulator terlebih dahulu.

Lab 07 - Analyze periodicity

Objective: Mengukur callback interval dari controller JSONL log.

```
python telemetry-analyze.py \
  module23-c2/logs/controller.jsonl
```

- Perhatikan count, median interval, min/max, dan coefficient of variation.

Verify: Lo bisa menjelaskan mengapa periodicity saja tidak cukup untuk verdict.

4.1 Task sequence reasoning

Timestamp	Event	Interpretasi
10:00:00	register	Agent identity muncul.
10:00:05	poll -> no task	Heartbeat / idle callback.
10:00:10	poll -> system_info	Task diambil.
10:00:10	result	Structured output kembali.
10:00:15	poll -> no task	Kembali idle.

Defender sebaiknya mengorelasikan sequence seperti ini dengan process/network telemetry. C2 detection lebih kuat jika state transition terlihat, bukan hanya satu connection event.

4.2 Harmless artifact transfer

Lab 08 - Transfer a fixed training artifact

Objective: Memahami T1105 dari sisi observability tanpa arbitrary download.

```
python operator.py get_training_artifact \
  --agent lab-agent-01
```

- Agent hanya boleh mengambil /artifact/training.txt dari controller yang sama.

- Periksa file di module23-c2/artifacts/agent/ dan SHA-256 yang dicatat agent.

Verify: Network request, file creation, dan hash semuanya bisa dikorelasikan.

Lab note - Artifact fixed dan harmless; agent menolak URL/file path arbitrer.

5. Detection Engineering Perspective

Tujuan Red Team yang matang bukan hanya mendapatkan channel, tetapi memahami bukti apa yang muncul dan apakah kontrol pertahanan mendeteksinya. MITRE T1071 memiliki detection strategy yang menyoroti abnormal process-to-protocol mapping, unusual outbound bytes, dan beacon-like interval. Sysmon Event ID 3 menghubungkan network connection dengan ProcessGuid; Event ID 22 menghubungkan DNS query dengan process.

5.1 Detection hypothesis worksheet

Hypothesis	Telemetry minimum	False-positive questions
Unknown process makes periodic outbound connections.	ProcessCreate + NetworkConnect + destination context.	Updater? EDR agent? Monitoring daemon?
Process repeatedly queries rare domain then connects.	DNS Query + NetworkConnect + process identity.	CDN rotation? Browser tab? Package manager?
Network transfer followed by new file and execution.	Network + FileCreate + hash + ProcessCreate.	Legit software install/update?
Service account host talks to unusual external endpoint.	Asset role + process + network baseline.	Expected vendor integration? Backup?

Lab 09 - Build a local detection hypothesis

Objective: Menulis analytic yang bisa diuji dengan simulator.

- Gunakan template notes/detection-hypothesis.md.

- Definisikan data source, entity join key, threshold/sequence, expected true positive, dan top 3 false positives.

- Jangan mulai dari IOC seperti 127.0.0.1:8230; mulai dari behavior pattern.

Verify: Hypothesis tetap masuk akal jika port/destination diubah.

5.2 Windows telemetry

Lab 10 - Review Sysmon network/DNS/process events

Objective: Menghubungkan process dan network activity pada Windows lab.

```
powershell -ExecutionPolicy Bypass -File .\windows-sysmon-review.ps1
```

- Script hanya membaca event log; tidak mengubah konfigurasi Sysmon.

- Jika Event ID 3/22 tidak tersedia, catat telemetry gap sebagai not-testable.

Verify: Output berisi ProcessGuid/ProcessId, Image, Destination/Query, timestamp jika telemetry ada.

Important - Sysmon Event ID 3 network connections disabled by default dan biasanya perlu di-enable secara selektif karena volume. Jangan menyimpulkan detection miss jika data source memang belum tersedia.

5.3 Linux telemetry

Lab 11 - Correlate process and socket

Objective: Membuktikan process yang memiliki koneksi simulator.

```
ss -tnp | grep 8230
ps -ef | grep -E "c2-(controller|agent)\.py"
```

- Opsional: inspect /proc/<pid>/cmdline dan /proc/<pid>/fd dalam lab milik sendiri.

Verify: Lo bisa menghubungkan PID agent/controller ke socket loopback.

6. Sliver Framework Literacy - Review Only

Sliver adalah cross-platform C2 framework untuk professional red teams dan dokumentasi resminya menyebut dukungan mTLS, WireGuard, HTTP(S), dan DNS dari satu operator console. Di module ini Sliver dipelajari sebagai reference implementation; hands-on berhenti di provenance verification, architecture/reference review, dan help output - tidak generate implant.

6.1 Supply-chain verification

Official Sliver documentation mempublikasikan Minisign public key untuk memverifikasi binary artifacts dari project authors. Karena workstation Red Team menjalankan tooling berprivilege tinggi, provenance check harus dianggap bagian security control, bukan formalitas.

Lab 12 - Verify a downloaded Sliver artifact

Objective: Memastikan binary dan signature berasal dari signing key resmi sebelum review.

- Download release + .minisig dari official release channel secara manual.

```
./sliver-verify.sh /path/to/sliver-server
```

- Helper memakai public key resmi Sliver dan tidak men-download atau execute binary.

Verify: minisign melaporkan signature valid sebelum file boleh dipakai.

Lab note - Binary signed oleh key resmi membuktikan provenance dari project authors, bukan menjamin binary bebas vulnerability atau cocok untuk scope lo.

6.2 Architecture review checklist

Area	Yang dipelajari	Yang sengaja belum dilakukan
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Server/operator	Separation operator identity, server state, multiplayer concept.	Membuka team server ke Internet.
Transports	mTLS, WireGuard, HTTP(S), DNS capability map.	Traffic encoder / evasion tuning.
Agent models	Session vs beacon concepts.	Generate/deploy implant.
Extensions	Armory/extension trust as supply-chain surface.	Load third-party offensive extensions.
Audit	Operational logs and operator attribution concepts.	Sensitive loot collection.

Lab 13 - Review Sliver help/reference

Objective: Membangun framework literacy tanpa creating payload.
/path/to/sliver-server --help
/path/to/sliver-server version
- Baca Architecture, Getting Started, transport references, dan audit/logging docs.
- Catat server state directories, operator identity model, dan transport choices di notes/sliver-review.md.
Verify: Tidak ada generated agent/implant di workspace.

Why no implant generation? - Karena learning objective module ini adalah C2 architecture, telemetry, provenance, dan detection. Agent generation/deployment menambah risiko dan tidak dibutuhkan untuk memahami konsep. Simulator lokal sudah memberi traffic/task/result yang bisa diamati.

6.3 Sliver and AV evasion

Dokumentasi Sliver sendiri menyatakan anti-virus evasion bukan scope utama project. Kurikulum kita juga memisahkan C2 fundamentals dari evasion research. Jadi jangan mengukur kualitas C2 dari apakah Defender/EDR berhasil dilewati; ukur architecture correctness, operational control, telemetry awareness, dan cleanup discipline.

7. Operator Hygiene & Safe Operations

Control	Praktik module ini	Kenapa
Bind address	127.0.0.1 untuk simulator.	Mencegah exposure ke LAN/Internet.
Task capability	Allow-list structured actions.	Menghindari arbitrary command execution.
Secrets	Tidak hard-code credential/token nyata.	Repo/module tetap aman dibagikan.
Audit	JSONL dengan timestamps dan agent/operator metadata.	Reproducibility dan attribution.
Artifacts	Fixed training file + SHA-256.	Transfer dapat diuji tanpa payload berbahaya.
Cleanup	PID/socket/file/evidence checklist.	Menghindari listener/agent tertinggal.

7.1 C2 security failures dari sisi operator

- Team server listening ke public interface tanpa kebutuhan dan firewall boundary yang jelas.
- Shared operator credential sehingga tidak ada attribution.
- Private keys/token tersimpan di Git atau shell history.
- Third-party extension/alias dijalankan tanpa provenance review.
- Evidence/loot dibiarkan world-readable atau tidak dienkripsi.
- Agent/listener tidak di-cleanup setelah exercise.
- Scope tidak memisahkan allowed hosts, allowed protocols, dan stop conditions.

7.2 Evidence and audit discipline

Lab 14 - Generate evidence manifest

Objective: Membuktikan integritas log, PCAP, result, dan artifact.
./evidence-manifest.sh module23-c2
sha256sum -c module23-c2/evidence/manifest.sha256
Verify: Semua evidence file menghasilkan OK sebelum archive.

Lab 15 - Cleanup verification

Objective: Memastikan tidak ada component lab tertinggal.
- Stop agent dan controller.
ss -ltnp | grep 8230 || true
pgrep -af "c2-(controller|agent)\.py" || true
- Review workspace; retain evidence yang diperlukan, hapus temporary artifact kalau exercise selesai.
Verify: Tidak ada listener/process lab tersisa.

7.3 What good reporting looks like

Section	Isi minimum
Objective	Apa yang diuji: C2 lifecycle, telemetry, detection, atau tooling provenance.
Topology	Operator/server/agent, transport, IP/port lab, scope.
Actions	Timestamp, actor/operator, task, target agent, result.
Telemetry	Process, network, DNS, file, PCAP/log references.
Detection result	blocked/detected/logged-only/missed/not-testable + alasan.
Cleanup	Listener/process/artifact removed, evidence archived.

8. Guided Capstone - Local C2 Detection Exercise

Scenario - Lo bertindak sebagai Red + Blue analyst untuk simulator localhost. Goal bukan menyembunyikan C2; goal adalah membangun satu C2

lifecycle yang terkontrol lalu membuktikan telemetry dan detection hypothesis secara reproducible.

Capstone sequence

- 1. Initialize workspace dan catat scope: localhost only, port 8230, no external destination.
- 2. Start controller dan satu training agent; record PID, socket, and registration evidence.
- 3. Capture loopback PCAP.
- 4. Queue system_info, process_summary, network_summary, dan satu echo task.
- 5. Queue fixed training artifact transfer dan verify SHA-256.
- 6. Stop PCAP; run telemetry-analyze.py.
- 7. Write one detection hypothesis based on process + periodic network behavior.
- 8. If Windows Sysmon lab is available, review analogous ProcessCreate/NetworkConnect/DNSQuery visibility; mark missing data not-testable.
- 9. Review Sliver official architecture and verify a downloaded server binary signature if desired; do not generate an implant.
- 10. Generate evidence manifest, stop components, verify cleanup, write final summary.

Required deliverables

Deliverable	Pass criteria
topology.md	Menjelaskan role/trust boundary dan localhost-only scope.
controller.jsonl	Register/poll/task/result sequence terlihat.
c2-local.pcap	Hanya traffic lab port 8230.
telemetry-summary.md	Process, destination, interval, bytes, artifact transfer.
detection-hypothesis.md	Behavior-based, data source jelas, false positives dibahas.
cleanup.md	No listener/process left; artifact retention decision documented.
manifest.sha256	Evidence integrity check passes.

Scoring rubric

Area	Excellent	Needs work
Architecture	Trust boundaries + failure modes jelas.	Sekadar gambar arrows.
Telemetry	Process/network/DNS/file correlated by time/entity.	Hanya screenshot satu event.
Detection	Behavior-based + false positives + data availability.	IOC-only rule.
Safety	Loopback-only, allow-list, cleanup verified.	Unknown listener/artifact tersisa.
Evidence	Hash + timestamps + reproducible steps.	Notes tidak bisa direproduce.

9. Troubleshooting

Problem	Check
Agent cannot register	Controller running? 127.0.0.1:8230 listener? Wrong workspace?
No task result	Agent still polling? Agent ID exact? Check controller JSONL.
tcpdump empty	Capture interface lo, correct port 8230, traffic generated after capture started.
Sysmon Event 3 absent	NetworkConnect may be disabled; do not call it a miss.
Sliver verification fails	Correct .minisig for exact binary? Official Minisign key? Binary modified?

10. Cheat Sheet

```
# local simulator
python c2-controller.py --workspace module23-c2
python c2-agent.py --workspace module23-c2 --agent-id lab-agent-01
python operator.py state
python operator.py system_info --agent lab-agent-01
python operator.py get_training_artifact --agent lab-agent-01

# observe
ss -ltnp | grep 8230
sudo tcpdump -i lo -nn tcp port 8230 -w module23-c2/captures/c2-local.pcap
python telemetry-analyze.py module23-c2/logs/controller.jsonl

# evidence / cleanup
./evidence-manifest.sh module23-c2
sha256sum -c module23-c2/evidence/manifest.sha256
pgrep -af 'c2-(controller|agent)\.py' || true

# review-only Sliver provenance
./sliver-verify.sh /path/to/sliver-server
```

11. Knowledge Check

Question	Expected idea
Kenapa C2 != remote shell?	Ada identity, task queue, transport, results, audit, artifact, lifecycle.
Kenapa HTTPS alone bukan indikator?	Normal traffic dominan HTTPS; perlu process/destination/timing context.
Apa beda session dan beacon?	Continuous/interactive vs periodic/asynchronous check-in.
Apa nilai Sysmon Event 3?	Mengaitkan network connection dengan process identity/ProcessGuid.
Kenapa IOC-only lemah?	IP/domain berubah; behavior relationship lebih tahan perubahan.
Apa tujuan signature verification?	Software provenance/integrity sebelum tool execution.

Kenapa module tidak generate Sliver implant?

Learning goal tercapai dengan simulator; menekan risiko dan scope creep.

11.1 Detection Logic - Pseudocode

Detection logic berikut sengaja berbentuk pseudocode agar mindset-nya portable ke SIEM mana pun. Fokusnya bukan signature port 8230, melainkan relationship antar-event.

```
network = group_by(process_image, destination)
for each group:
    intervals = delta(sorted(timestamps))
    if count >= 6 and low_variance(intervals):
        enrich(process_parent, signer, asset_role, destination_age)
        if unusual_process_network_pair and rare_destination:
            raise behavioral_hypothesis

# transfer sequence
if network_download -> file_create -> process_create within short_window:
    investigate provenance, signer, user context, and expected software workflow
```

Field untuk korelasi	Contoh	Nilai analitik
Process identity	Image, hash, signer, ProcessGuid	Membedakan browser/updater vs proses tidak lazim.
Network identity	Destination IP/domain/port/TLS SNI	Novelty, allow-list, asset relation.
Timing	first_seen, interval, duration	Periodicity dan sequence.
File activity	path, hash, creation time	Transfer -> artifact relationship.
User/host context	account, hostname, role	Apakah behavior sesuai fungsi host?

11.2 Minimum C2 Audit Log Schema

Field	Kenapa wajib
timestamp_utc	Urutan task/result dan korelasi dengan endpoint/network telemetry.
operator_id	Attribution dan accountability.
agent_id / host_id	Target identity yang stabil.
task_id + task_type	Menghubungkan queue, delivery, result, dan error.
transport / listener	Mengetahui channel yang dipakai.
result_status	success/error/timeout tanpa harus menyimpan secret output di log utama.
artifact_hash	Integrity reference untuk file yang ditransfer.
cleanup_status	Membuktikan teardown selesai.

12. References

MITRE ATT&CK - Command and Control tactic (TA0011): <https://attack.mitre.org/tactics/TA0011/>

MITRE ATT&CK - Application Layer Protocol (T1071): <https://attack.mitre.org/techniques/T1071/>

MITRE ATT&CK - Web Protocols (T1071.001): <https://attack.mitre.org/techniques/T1071/001/>

MITRE ATT&CK - Ingress Tool Transfer (T1105): <https://attack.mitre.org/techniques/T1105/>

Microsoft Sysmon events - ProcessCreate, NetworkConnect, DNS Query: <https://learn.microsoft.com/en-us/windows/security/operating-system-security/sysmon/sysmon-events>

Microsoft Sysinternals Sysmon: <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

Sliver official documentation: <https://sliver.sh/docs/>

Sliver official Minisign public key and verification: <https://sliver.sh/docs/?name=Minisign+Public+Key>

Arch Linux minisign package: https://archlinux.org/packages/extra/x86_64/minisign/

Arch Linux Wireshark package: https://archlinux.org/packages/extra/x86_64/wireshark-qt/

Exit criteria - Sebelum lanjut ke Module 24, lo harus mampu menggambar C2 lifecycle, menjalankan simulator localhost, membaca task/result sequence, menangkap traffic, mengukur periodicity, menulis detection hypothesis, menjelaskan Sysmon Event 1/3/22 relevance, memverifikasi provenance Sliver artifact, dan membuktikan cleanup.